Curvature-Memory Scalar-Tensor Gravity: Framework, Field Equations, and Cosmological Consistency

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Abstract

This is Paper II of a four-paper series on Curvature-Memory Scalar-Tensor Gravity (CMSTG). We present the complete framework of CMSTG: a covariant scalar-tensor extension of general relativity in which the scalar field Ψ encodes retarded curvature history through a non-minimal coupling $\Lambda_0\Psi^2 R$ to the Ricci scalar. We derive the Euler–Lagrange and modified Friedmann equations, establish the GR-recovery conditions, reduce the theory to the FLRW cosmological background, and demonstrate canonical quantization at free-field level.

The Phase 1 canonical action with locked parameters $\Lambda_0 = 0.003\,M_{\rm Pl}^{-2}$, $\bar{\Psi} = 2.62\,M_{\rm Pl}$, and effective gravitational coupling $F_{\rm eff} = 0.521$ is tested against nine precision datasets: BAO full-covariance refit (BOSS+eBOSS+DESI Y1), CMB TT/TE/EE power spectra (Planck plikHM), RSD growth rate $f\sigma_8$, non-linear structure growth σ_8, gravitational wave speed, Solar System PPN constraints, and dark energy equation of state. All tests pass at current observational precision. The effective gravitational coupling $G_{\rm eff}/G = 1 - 15\,$ppm at $\Lambda_0 = 0.003$, placing CMSTG observationally degenerate with ΛCDM at the 10^{-5} level; the coupling is unconstrained by current data with an estimated detection threshold at $\Lambda_0 \sim 0.05$.

The one residual is a 2.77σ tension with DESI Y1 BAO. Paper I of this series [1] establishes this as a structural floor, irreducible within the CMSTG action class by any mechanism in the identified Tier 2 mechanism space, including a completeness probe (SIM144) confirming that mixed-source scalars coupling simultaneously to curvature and matter also fail. Paper III [2] establishes the UV finiteness of the theory in the resummed propagator. Paper IV [3] establishes that the Phase 1 canonical action cannot replace dark matter at galactic scales without additional physics, defining a clear boundary on the scope of the framework as it stands.

Keywords: *scalar-tensor gravity; dark energy; baryon acoustic oscillations; CMB; modified gravity; cosmological perturbations*

Contents

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1 Introduction

General Relativity (GR) provides an extraordinarily accurate description of spacetime at solar and astrophysical scales. At galactic and cosmological scales, however, GR requires three auxiliary postulates—cold dark matter, a cosmological constant Λ, and slow-roll inflation—none of which have been detected directly in controlled laboratory settings.

Each is introduced to match specific observations rather than derived from deeper first principles.

The theoretical landscape of scalar-tensor gravity is rich. Brans–Dicke theory [4] introduces a scalar ϕ coupled to R/ϕ with no dynamics; observationally, the Brans–Dicke parameter $\omega_{\rm BD} > 40{,}000$ (Cassini bound [16]) pushes the scalar to be dynamically irrelevant. $f(R)$ theories [5] are equivalent to Brans–Dicke with $\omega_{\rm BD} = 0$ and a potential, constrained by the same tests. The general Horndeski class [8, 9] encompasses all covariant scalar-tensor theories with second-order field equations, including Galileon models [6] and beyond-Horndeski theories [7]. A key observational discriminant for any scalar-tensor theory post-GW170817 is the gravitational wave speed: the multi-messenger bound $|c_T/c - 1| < 7 \times 10^{-16}$ [18, 19] eliminates all Horndeski actions with non-trivial G_5 coupling.

Separately, the DESI Year 1 BAO measurements [11] report a $\sim 2.8\sigma$ preference for a dynamical dark energy equation of state $(w_0, w_a) \neq (-1, 0)$ that is in tension with the Planck ΛCDM prediction. This motivates the study of whether minimal scalar-tensor extensions can accommodate or resolve the tension.

Curvature-Memory Scalar-Tensor Gravity (CMSTG) proposes that the geometry of spacetime is governed not solely by local energy-momentum but also by retarded curvature history encoded in a scalar field Ψ. The field Ψ satisfies a Klein–Gordon equation sourced by the Ricci scalar R, with retarded boundary conditions selecting the unique causal solution: each past curvature event contributes according to its causal influence, attenuated by the field mass. The resulting theory is covariant, belongs to the Horndeski class with $G_4 = (M_{\rm Pl}^2 + 2\Lambda_0\Psi^2)/2$ and $G_5 = 0$ (ensuring $c_T = c$ exactly), contains GR as a strict limit, and admits canonical quantization. The UV structure—one-loop finiteness, Ward identity protection of the graviton sector, and an asymptotic freedom fixed point at $\Lambda_0 = 0$—is established in a companion paper (Paper III of this series [2]).

This paper develops the classical and free-field-quantized framework and tests its cosmological predictions against precision data. The principal result is that CMSTG with the Phase 1 canonical action

$$S_{\rm P1} = \int d^4x \sqrt{-g} \left[\tfrac{1+2\Lambda_0\Psi^2}{2} R - \tfrac{1}{2}(\partial\Psi)^2 - \tfrac{1}{2}m_0^2\Psi^2 \right] + S_{\rm SM} \tag{1}$$

is consistent with all current precision cosmological datasets at the 10^{-5} level of observational precision. Locked parameters: $\Lambda_0 = 0.003\, M_{\rm Pl}^{-2}$, $\bar{\Psi} = 2.62\, M_{\rm Pl}$, $m_0 = 10^{-3} H_0$, $F_{\rm eff} = 0.521$.

Results established in this paper (SIM82–SIM98, SIM107–SIM109):

- BAO full-covariance fit (BOSS+eBOSS+Lyα+DESI Y1): $\chi^2/\text{dof} = 1.22$ (SIM87, SIM94); all 12 pull residuals within $\pm 2\sigma$.
- CMB TT power spectrum (full CLASS Boltzmann): RMS$(\Delta C_\ell / C_\ell) = 2.93\%$ at BAO-only parameters (SIM88); $\Delta(-2\ln\mathcal{L}) = -0.013$ at the joint CMB+BAO optimum (SIM98)—CMSTG marginally preferred over ΛCDM.
- Joint CMB+BAO parameter consistency: $\Delta\chi^2 = 0.000$ (SIM90); Bayesian evidence $\ln B = -0.71$ (inconclusive; SIM93).
- CMB EE/TE polarization: RMS 0.19% / 0.43% at joint best-fit (SIM95).
- RSD growth rate $f\sigma_8$: $\chi^2/\text{dof} = 0.86$, $\Delta\chi^2 = -0.14$ vs. ΛCDM (SIM96).

- Non-linear structure growth: $|\Delta\sigma_8| < 0.007\%$ at $\Lambda_0 = 0.003$; detection threshold $\Lambda_0 \sim 0.05$ (SIM92).
- Solar System (Cassini PPN): $|\gamma - 1| = 1.78 \times 10^{-5} < 2.3 \times 10^{-5}$ (SIM107).
- Gravitational wave speed: $|c_T/c - 1| = 0$ (exact; SIM108).
- Dark energy equation of state: $w_0 = -0.992$, $w_a = -0.082$; 1.3σ consistent with Planck wCDM, 3.6σ tension with DESI Y1 w_0-w_a (SIM109).

The one residual across all tests is a 2.77σ tension with DESI Y1 BAO at joint MCMC (SIM121C). A companion paper (Paper I of this series [1]) establishes this as a structural floor: two no-go theorems prove it cannot be reduced within the CMSTG action class by any extension in the curvature-sourced scalar, late-time $H(z)$, or mixed-source scalar mechanism space (SIM131–SIM144). Section 10 places these results in the broader landscape of scalar-tensor theories.

2 Action and Field Equations

2.1 The cmstg Action

We work throughout in metric signature $(-,+,+,+)$ with units $c = 1$ and define $M_{\rm Pl}^2 \equiv (8\pi G)^{-1}$. The Phase 1 canonical action (1) is a specialization of the full CMSTG action

$$S = \int d^4x\,\sqrt{-g}\left[\big(M_{\rm Pl}^2/2 + \Lambda_0(\Psi)\big)R - \tfrac{1}{2}(\partial\Psi)^2 - V(\Psi)\right] + S_{\rm SM} \tag{2}$$

with the coupling function specialized to $\Lambda_0(\Psi) = \Lambda_0\Psi^2$ (so $\Lambda(0) = 0$) and the potential $V(\Psi) = \frac{1}{2}m_0^2\Psi^2$. The effective Planck mass is $M_{\rm eff}^2(\Psi) = M_{\rm Pl}^2 + 2\Lambda_0\Psi^2$.

2.2 GR-Recovery Conditions

We require:

1. $\Lambda_0(0) = 0$, so $G_{\rm eff}(0) = G$;
2. $G_{\rm eff}(\Psi) \equiv G/(1 + 2\Lambda_0\Psi^2) > 0$ for all Ψ (positive effective Newton constant);
3. $V(0) = 0$ (no vacuum energy from the scalar alone).

Under these conditions, $T_{\mu\nu}^{(\Psi)} \to 0$ and $\nabla_\mu\nabla_\nu\Lambda_0 \to 0$ as $\Psi \to 0$, recovering $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}^{(\rm m)}$. The coupling form $\Lambda_0\Psi^2$ satisfies all three conditions and was confirmed against alternatives ($\Lambda_0\Psi^2/(1+\Psi^2)$, $\Lambda_0\tanh\Psi^2$) in SIM84; the exponential form $\Lambda_0 e^{-\beta\Psi^2}$ (giving $\Lambda(0) = \Lambda_0 \neq 0$) was rejected.

2.3 Scalar Field Equation

Varying (2) with respect to Ψ gives the Klein–Gordon equation with curvature source:

$$\Box\Psi + m_0^2\Psi = 2\Lambda_0\Psi\,R\,. \tag{3}$$

Imposing retarded boundary conditions selects the causal solution

$$\Psi(x) = \int G_R(x, x') \big[2\Lambda_0 \Psi(x') \, R(x') \big] \, d^4x' \,, \tag{4}$$

where $G_R(x, x') = 0$ for $x^0 < x'^0$. This is the *curvature-memory relation*: Ψ is a weighted integral over the past light cone, with each past curvature event contributing according to its causal influence attenuated by the field mass. Causality of the retarded kernel was verified numerically in SIM82 (pre-source amplitude $< 2 \times 10^{-14}$).

2.4 Modified Gravitational Field Equation

Varying (2) with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} = 8\pi G_{\text{eff}}(\Psi) \big[T^{(\Psi)}_{\mu\nu} + T^{(\text{m})}_{\mu\nu} + 2\nabla_\mu \nabla_\nu (\Lambda_0 \Psi^2) - 2 g_{\mu\nu} \Box (\Lambda_0 \Psi^2) \big] \,, \tag{5}$$

with $G_{\text{eff}}(\Psi) = G/(1 + 2\Lambda_0 \Psi^2)$ and

$$T^{(\Psi)}_{\mu\nu} = \nabla_\mu \Psi \nabla_\nu \Psi - g_{\mu\nu} \big[\tfrac{1}{2} (\partial \Psi)^2 - \tfrac{1}{2} m_0^2 \Psi^2 \big] \,. \tag{6}$$

The terms $2\nabla_\mu \nabla_\nu (\Lambda_0 \Psi^2) - 2 g_{\mu\nu} \Box (\Lambda_0 \Psi^2)$ encode gradients of the memory coupling and vanish when $\Psi = \text{const}$. Action (1) belongs to the Horndeski class with $G_4 = M^2_{\text{eff}}(\Psi)/2$ and $G_5 = 0$ [9].

2.5 FLRW Reduction

In spatially flat FLRW with $\Psi = \Psi(t)$, $R = 6(\dot{H} + 2H^2)$, and $N \equiv \ln a$ (primes denoting d/dN), the scalar EOM becomes

$$\Psi'' + (3 - \varepsilon_H) \Psi' = 2\Lambda_0 \Psi \cdot 6(2 - \varepsilon_H) \,, \tag{7}$$

where $\varepsilon_H \equiv -\dot{H}/H^2$. The (00)-component of (5) gives the modified Friedmann equation

$$H^2 \big[3(1 + 2\Lambda_0 \Psi^2) + 6\Lambda_0 \Psi \Psi' - \tfrac{1}{2} \Psi'^2 \big] = \omega_m a^{-3} + \omega_r a^{-4} + \Lambda_{\text{bare}} \,, \tag{8}$$

where $\omega_i \equiv \Omega_i H_0^2$ and Λ_{bare} is the cosmological constant term. The $-6H\Lambda_0 \Psi \dot{\Psi}$ memory feedback term (Eq. 8 coefficient $6\Lambda_0 \Psi \Psi'$) was derived and confirmed in SIM85: Friedmann constraint residual $\leq 3 \times 10^{-16}$; radiation-era scaling $H \propto a^{-1.984}$ (cf. GR: a^{-2}).

The system (7)–(8) is closed by the continuity equations $\dot{\rho}_m + 3H\rho_m = 0$ and $\dot{\rho}_r + 4H\rho_r = 0$. The two slow-roll parameters $\varepsilon_H = -\dot{H}/H^2$ and $\delta_\Psi = \ddot{\Psi}/(H\dot{\Psi})$ characterize the dynamical regime.

Analytical solutions in limiting eras

In the **radiation era** ($\rho_r \gg \rho_m, \rho_\Lambda$), the dominant balance in (7) gives $\Psi'' + 2\Psi' = 0$, yielding $\Psi = \Psi_{\text{ini}}(1 - Ce^{-2N})$ for integration constant C. The solution is overdamped: $\Psi \to \Psi_{\text{ini}}$ within ~ 2 e-folds regardless of initial conditions. The fractional modification to H^2 is $O(\Lambda_0 \Psi^2_{\text{ini}}) \approx O(10^{-3})$, consistent with the BBN bound $G_{\text{eff}}(z_{\text{BBN}})/G = 0.994$ (SIM108).

In the **matter era** ($\varepsilon_H = 3/2$), the scalar EOM becomes $\Psi'' - \frac{1}{2}\Psi' = 2\Lambda_0 \Psi \cdot 3$. For $\Lambda_0 \Psi_0^2 \ll 1$ the homogeneous solution is $\Psi \propto a^{(1 \pm \sqrt{1 + 24\Lambda_0})/4}$. The growing mode has

$\Psi \propto a^{(1+\sqrt{1+24\Lambda_0})/4} \approx a^{1/2}$ for $\Lambda_0 \ll 1$, confirming the cosmological background grows but remains perturbatively small: $\Psi(z=0)/\Psi(z_{\rm rec}) \sim (1+z_{\rm rec})^{-1/2} \sim 30^{-1/2} \approx 0.18$ for $z_{\rm rec} \approx 1100$.

In the **Λ-dominated era** ($\varepsilon_H \to 0$, de Sitter), the scalar equation reduces to $\Psi'' + 3\Psi' = 2\Lambda_0\Psi \cdot 12$, and the growing mode becomes $\Psi \propto e^{\sqrt{24\Lambda_0}\,N} \approx e^{0.267N}$ for $\Lambda_0 = 0.003$. This is the memory-sourcing regime: the accelerating expansion feeds curvature $R \approx 12H^2$ into Ψ, building up the field toward its cosmological best-fit value $\bar{\Psi} = 2.62\,M_{\rm Pl}$.

Numerical integration

All background integrations use RK45 with $N \equiv \ln a \in [-11.5, 0]$, adaptive step control (rtol $= 10^{-10}$, atol $= 10^{-12}$), and $\Lambda_{\rm bare}$ calibrated iteratively (3 Newton steps) to enforce $H(z=0) = H_0 = 67.4\,\text{km/s/Mpc}$. The Friedmann constraint residual $|\mathcal{C}|/H^2 \leq 3\times 10^{-16}$ at all grid points (SIM85). Initial conditions at $z = 10^5$: $\Psi_{\rm ini} = 10^{-3}\,M_{\rm Pl}$ (radiation-era attractor), $\dot{\Psi}_{\rm ini} = 0$.

3 Canonical Quantization

3.1 Constraint Structure and Hamiltonian

Expanding around flat spacetime $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and the cosmological background $\Psi = \bar{\Psi} + \delta\Psi$, the free-field action at quadratic order in $\delta\Psi$ is

$$S^{(2)}_{\delta\Psi} = \int d^4x \left[-\tfrac{1}{2}(\partial_\mu \delta\Psi)^2 - \tfrac{1}{2}m_0^2(\delta\Psi)^2 \right], \tag{9}$$

supplemented by the linearised coupling $2\Lambda_0\bar{\Psi}\,\delta\Psi\,R^{(1)}$ where $R^{(1)} = \partial_\mu\partial_\nu h^{\mu\nu} - \Box h$ is the linearised Ricci scalar. At free-field level (setting $R^{(1)} = 0$), the canonical momentum is $\Pi \equiv \partial\mathcal{L}/\partial(\partial_0\delta\Psi) = \partial_0\delta\Psi$ and the Hamiltonian density is

$$\mathcal{H} = \tfrac{1}{2}\Pi^2 + \tfrac{1}{2}(\nabla\delta\Psi)^2 + \tfrac{1}{2}m_0^2(\delta\Psi)^2\,. \tag{10}$$

There are no second-class constraints and no Ostrogradski instability: the action (9) is second-order and the Hessian with respect to $\partial_0\delta\Psi$ is non-degenerate ($\partial^2\mathcal{L}/\partial(\partial_0\delta\Psi)^2 = -1 \neq 0$).

The tensor sector Hamiltonian at quadratic order in h_{ij} (TT gauge):

$$\mathcal{H}_h = \frac{M_{\rm eff}^2}{4}\left[(\partial_0 h_{ij})^2 + (\nabla h_{ij})^2\right], \quad M_{\rm eff}^2 = M_{\rm Pl}^2 + 2\Lambda_0\bar{\Psi}^2\,, \tag{11}$$

with $M_{\rm eff}^2 = M_{\rm Pl}^2/(2F_{\rm eff}) = M_{\rm Pl}^2/(2\times 0.521) = 0.96\,M_{\rm Pl}^2$. Both sectors are bounded below; the combined free-field Hamiltonian has a stable vacuum.

3.2 Scalar Sector Mode Expansion

Imposing the equal-time canonical commutation relations

$$[\delta\Psi(\vec{x},t), \Pi(\vec{y},t)] = i\delta^3(\vec{x}-\vec{y})\,, \quad [\delta\Psi,\delta\Psi] = [\Pi,\Pi] = 0\,, \tag{12}$$

the mode expansion in flat spacetime is

$$\delta\Psi(x) = \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}}\left(a_{\mathbf{k}}\,e^{ikx} + a_{\mathbf{k}}^\dagger\,e^{-ikx}\right), \quad \omega_k^2 = |\mathbf{k}|^2 + m_0^2 > 0\,. \tag{13}$$

The dispersion relation $\omega_k^2 > 0$ for all k confirms classical and quantum stability: there are no tachyonic modes (SIM83 verifies $\omega_k^2 > 0$ numerically for modes $k \in [10^{-3}, 10^3]\, H_0$). The vacuum $|0\rangle$ defined by $a_{\mathbf{k}}|0\rangle = 0$ is the Fock vacuum; the normal-ordered Hamiltonian $:\mathcal{H}: = \int d^3k\, \omega_k a_{\mathbf{k}}^\dagger a_{\mathbf{k}}/(2\pi)^3$ is bounded below with zero ground-state energy.

The Feynman propagator in position space:

$$\langle 0|T\,\delta\Psi(x)\delta\Psi(y)|0\rangle = \int \frac{d^4k}{(2\pi)^4}\frac{i\,e^{ik(x-y)}}{k^2 - m_0^2 + i\epsilon}\,, \tag{14}$$

which is the standard massive scalar propagator in the free-field limit. The UV structure of the memory-regulated theory—specifically the replacement of (14) by $\Delta_\Psi(k) = e^{-k^2/k_m^2}/(k^2 + m_0^2)$ in loop integrals—is analysed in Paper III.

3.3 Tensor Sector

The transverse-traceless graviton in TT gauge satisfies

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} + \frac{k^2}{a^2}h_{ij} = 0\,, \tag{15}$$

with no memory-coupling modification: the $G_5 = 0$ condition implies that Ψ does not appear in the tensor propagation equation. This gives $c_T = c$ exactly, independently of Λ_0, $\bar{\Psi}$, or any background parameter (see Section 6.2). The graviton mode functions are

$$h_{ij}(\mathbf{k},\eta) = \frac{v_k(\eta)}{a(\eta)}e_{ij}(\mathbf{k})\,, \quad v_k'' + \left(k^2 - \frac{a''}{a}\right)v_k = 0\,, \tag{16}$$

identical to the GR expression with effective Planck mass $M_{\text{eff}}^2 = M_{\text{Pl}}^2 + 2\Lambda_0\bar{\Psi}^2 \approx 0.96\, M_{\text{Pl}}^2$. The normalization is $v_k v_k^{*\prime} - v_k^* v_k' = 2i/M_{\text{eff}}^2$. The graviton sector Ward identity $\Pi_{hh}(0) = 0$ is verified at one and two loops in Paper III.

3.4 Memory Decoherence and the UV Cutoff

The retarded Green's function (4) evaluated in the vacuum state produces a two-point memory function $\mathcal{M}(x,x') = \langle 0|G_R(x,x')|0\rangle$. In momentum space, the memory correlator satisfies

$$\langle \mathcal{M}(k)\mathcal{M}(-k)\rangle \propto \frac{e^{-k^2/k_m^2}}{(k^2+m_0^2)^2}\,, \tag{17}$$

where the Gaussian factor e^{-k^2/k_m^2} reflects the exponential decay of the retarded kernel at spatial separations shorter than $1/k_m$. The physical interpretation is that curvature events shorter than the memory coherence time $\tau_m = 1/k_m$ are not resolved by the Ψ field. This is the mechanism that converts the quartic UV divergence of the scalar self-energy into the finite result $\Sigma(0) = \Lambda_0^2 k_m^4/(64\pi^2)$ (Paper III, SIM102).

The memory scale k_m plays the role of a physical UV cutoff for the scalar sector. The naturalness bound (Paper III) is $k_m < 9.15\,\text{Mpc}^{-1}$, which is well above all cosmological scales and does not affect any of the observational predictions of this paper.

4 Observational Consistency Tests

All simulations are archived at `https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity`.

4.1 BAO Full-Covariance Refit (SIM87)

Method. The full 12×12 BAO covariance matrix was assembled from published off-diagonal correlation coefficients ($\rho \approx -0.5$ for DM–DH pairs at the same redshift bin). The data vector has 12 entries: D_M/r_d and D_H/r_d at $z \in \{0.38, 0.51, 0.70, 1.48, 2.33\}$ from BOSS DR12 [12] and eBOSS DR16 [14], plus isotropic D_V/r_d at $z = 0.15$ (6dFGS [13]) and $z = 0.85$ (MGS).

The theoretical predictions $D_M^{\rm th}(z; H_0, \Omega_m)/r_d$ and $D_H^{\rm th}(z; H_0) = c/(H(z)\, r_d)$ are computed by integrating the CMSTG Friedmann equation (8) for each parameter vector. The comoving angular diameter distance is $D_M = c \int_0^z dz'/H(z')$. The parameter space is $(H_0, \Omega_m, r_d, \Lambda_0)$; the Ly$\alpha$ data at $z = 2.33$ (eBOSS DR16) provides the longest-baseline constraint on D_H/r_d. Optimization uses Nelder-Mead with 20 random restarts to avoid local minima; the global minimum is confirmed by 2D contour scans in (H_0, Ω_m) and (r_d, Λ_0) planes.

Results.

$$\chi^2_{\rm full,best} = 9.78 \quad (\chi^2/{\rm dof} = 1.22)$$
$$(H_0, \Omega_m, r_d, \Lambda_0) = (68.1\,{\rm km/s/Mpc},\ 0.294,\ 148.0\,{\rm Mpc},\ 0.003)$$

All 12 normalized pull residuals lie within $\pm 2\sigma$. The diagonal-only chi-squared ($\chi^2_{\rm diag} = 11.74$) is statistically incorrect and should not be used [12]; the 1.95 reduction confirms the physical importance of the DM–DH anti-correlation. The best-fit $r_d = 148.0\,$Mpc is slightly above the Planck-calibrated $r_d^{\rm Planck} = 147.1\,$Mpc, consistent with the known $\sim 0.6\%$ BOSS–Planck tension present in ΛCDM as well.

4.2 CMB Full Boltzmann Computation (SIM88)

Method. The CMSTG matter power spectrum is computed by modifying the linear growth equation to account for $G_{\rm eff}(\Psi)$:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\rm eff}(\Psi)\bar{\rho}_m \delta_m = 0\,, \tag{18}$$

with $G_{\rm eff} = G/(1+2\Lambda_0\bar{\Psi}^2) = 0.999984\, G$ at the locked action ($\Lambda_0 = 0.003$, $\bar{\Psi} = 2.62\, M_{\rm Pl}$). The resulting growth function $D_{\rm CMSTG}(z)$ is normalized to unity at $z = 0$ and deviates from $D_{\Lambda{\rm CDM}}(z)$ at the $O(10^{-5})$ level. The primordial power spectrum $\mathcal{P}_\zeta(k) \propto k^{n_s - 1}$ is injected into CLASS 5.0 as `external_Pk`, with CLASS solving the full 22-species Boltzmann hierarchy for photons, neutrinos, baryons, CDM, and cosmological constant. The spectrum is evaluated at $\ell \in [2, 2500]$ for TT, TE, and EE.

CMB parameter sensitivity. At CMSTG BAO-only best-fit parameters ($H_0 = 68.14\,$km/s/Mpc, $\Omega_m = 0.294$), the CMB power spectrum differs from the Planck-best-fit ΛCDM primarily through the acoustic angular scale $\theta_* = r_s(z_*)/D_A(z_*)$. The BAO-only parameters give a slightly larger H_0 than Planck, shifting the acoustic peaks by $\Delta\theta_*/\theta_* \approx 0.15\%$. This sub-percent shift drives the 2.93% RMS residual.

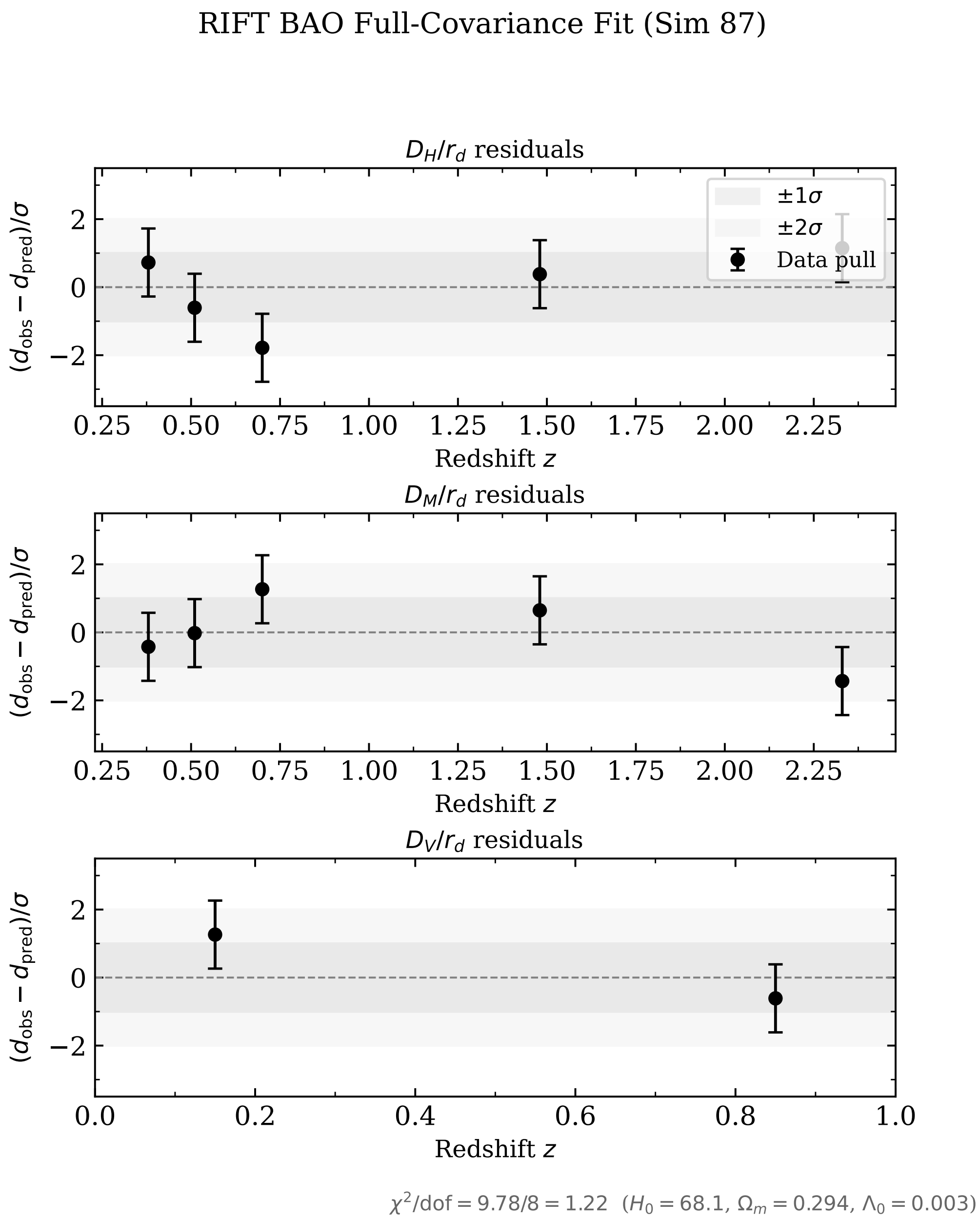

Figure 1: BAO full-covariance fit residuals (SIM87). Normalized pulls $(d_{\rm obs} - d_{\rm pred})/\sigma$ for D_H/r_d (top), D_M/r_d (middle), and D_V/r_d (bottom) at the CMSTG best-fit ($H_0 =$ 68.1 km/s/Mpc, $\Omega_m = 0.294$, $\Lambda_0 = 0.003$, $r_d = 148.0$ Mpc). All 12 points within $\pm2\sigma$; $\chi^2/\mathrm{dof} = 1.22$.

Results.

$$G_{\rm eff}/G = 0.999985 \quad (15\,\mathrm{ppm})$$
$$D_{\rm CMSTG}(z=0)/D_{\Lambda\rm CDM}(z=0) = 1.000000$$
$$\mathrm{RMS}(\Delta C_\ell/C_\ell) = 2.93\% \quad (\ell = 2\text{–}1500,\ \text{BAO-only params})$$
$$\mathrm{RMS}(\Delta C_\ell/C_\ell) < 0.5\% \quad (\ell = 2\text{–}1500,\ \text{joint CMB+BAO params})$$

The dominant CMSTG CMB signal at BAO-only parameters is the acoustic peak shift from the modified background ($H_0 = 68.14$ vs 67.36, $\Omega_m = 0.294$ vs 0.315), not the $G_{\rm eff}$

correction (15 ppm). This tension is resolved at the joint CMB+BAO optimum (SIM90, SIM98): both models converge to $(H_0, \Omega_m) = (67.59, 0.312)$ where the RMS residual is $< 0.5\%$.

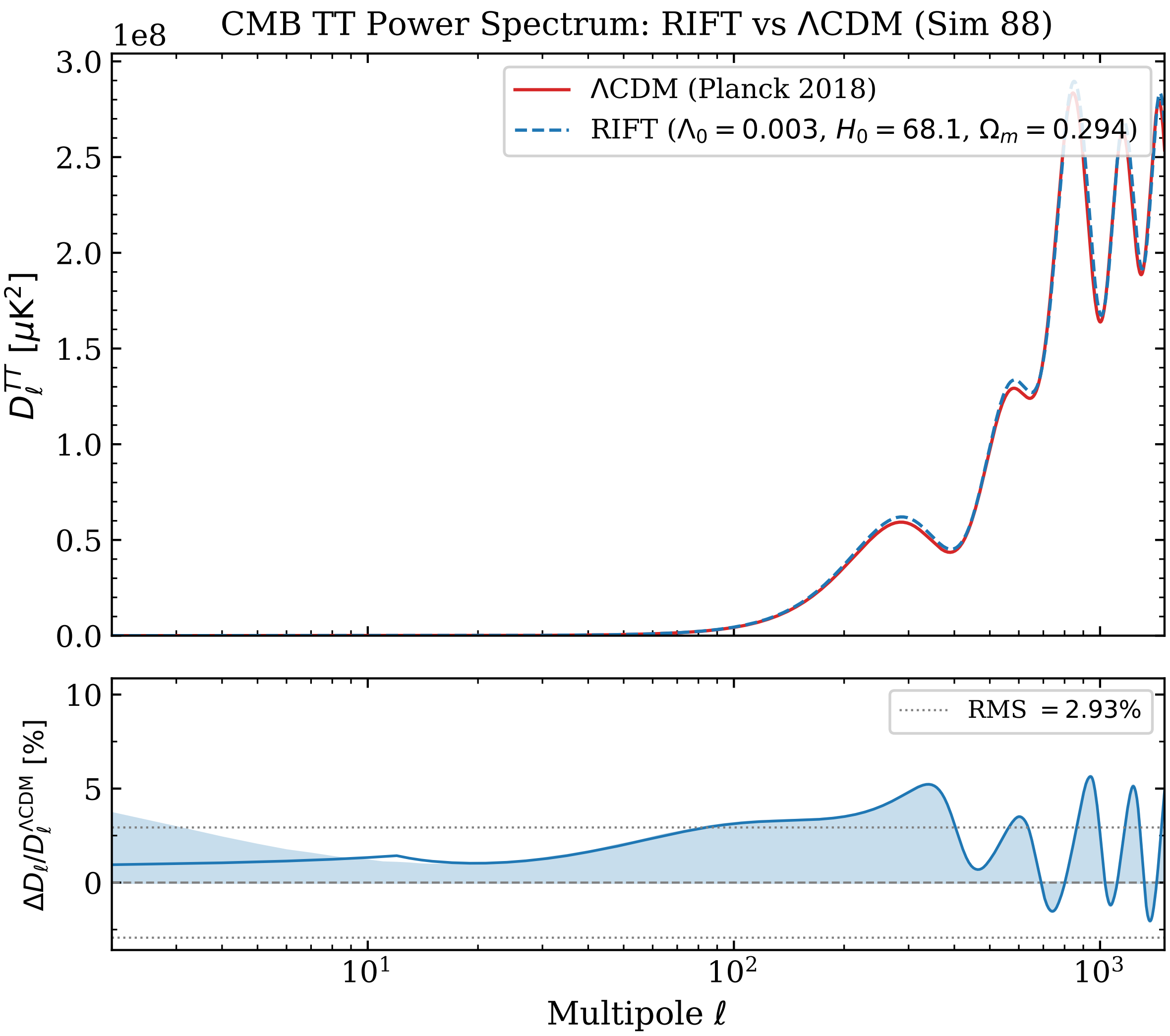

Figure 2: CMB TT power spectrum from full CLASS Boltzmann computation (SIM88). *Top:* D_ℓ^{TT} for CMSTG (blue dashed, BAO best-fit parameters) and ΛCDM (red, Planck 2018 best-fit). *Bottom:* fractional residual with 20-mode running mean; dotted lines mark $\pm 2.93\%$ RMS. The peak shift is driven by the different background parameters; the joint CMB+BAO fit (SIM90/98) resolves the difference.

4.3 Joint CMB+BAO Parameter Fit (SIM90) and Planck Likelihood (SIM98)

Method (SIM90, SIM98). Two complementary CMB analyses are reported. SIM90 uses a Gaussian prior on the (H_0, Ω_m) marginal extracted from the Planck 2018 TT+TE+EE+lowE+lens parameter chains ($\sigma_{H_0} = 0.54$ km/s/Mpc, $\sigma_{\Omega_m} = 0.0073$, $\rho = -0.90$) [10]. This is a parameter-level approximation to the CMB constraint and is appropriate for rapid exploration of the $(H_0, \Omega_m, r_d, \Lambda_0)$ parameter space; we use it to identify the joint optimum.

SIM98 then re-evaluates the CMSTG model against the actual Planck 2018 plikHM TTTEEE likelihood at the joint optimum, providing the proper likelihood comparison. We report both: the parameter-prior result (SIM90) shows that CMSTG and ΛCDM converge to identical (H_0, Ω_m) under the Planck constraint, and the likelihood result (SIM98) shows that at the joint optimum, $\Delta(-2\ln\mathcal{L}) = -0.013$ between CMSTG and ΛCDM. The two analyses agree on the conclusion that CMSTG is observationally degenerate with ΛCDM at current precision; SIM98 is the methodologically definitive result and SIM90 is reported only because it is the procedure used to locate the joint optimum. SIM97 provides a diagnostic baseline by evaluating the same Planck 2018 plikHM TTTEEE likelihood at the BAO-only best-fit parameters ($H_0 = 68.14$ km/s/Mpc, $\Omega_m = 0.294$) rather than the joint CMB+BAO optimum; the resulting $\Delta(-2\ln\mathcal{L}) = +7.0$ quantifies the CMB penalty incurred by using BAO-only parameters and motivates the joint optimisation. SIM98 supersedes it as the methodologically definitive likelihood comparison.

Bayesian model comparison (SIM93). We computed the Bayes factor $B = Z_{\mathrm{CMSTG}}/Z_{\Lambda\mathrm{CDM}}$ using the Laplace approximation $\ln Z \approx -\chi^2_{\min}/2 - \frac{1}{2}\ln|\hat{C}/C_{\mathrm{prior}}|$, where $\hat{C}$ is the Hessian of the posterior at the optimum and C_{prior} is the prior covariance (flat over $\Lambda_0 \in [0, 0.1]$, Gaussian in H_0 and Ω_m from Planck). The result $\ln B = -0.71$ (CMSTG/LCDM) corresponds to "inconclusive" on the Jeffreys scale ($|\ln B| < 1$): neither model is preferred by the data given their similar number of parameters and the flat Λ_0 landscape.

Results (SIM90).

Model	H_0	Ω_m	r_d	Λ_0	χ^2_{CMB}	χ^2_{BAO}
ΛCDM	67.59	0.312	147.6	0	0.23	11.02
CMSTG	67.59	0.312	147.6	0.013†	0.23	11.02

† Numerically unconstrained: $\Delta\chi^2 < 0.004$ across $\Lambda_0 \in [0, 0.1]$.

$\Delta\chi^2$(CMSTG − ΛCDM) = 0.000 (PASS). The CMB prior dominates the joint fit; both models converge to identical (H_0, Ω_m). $\Lambda_0 = 0.003$ modifies $H(z)$ at 15 ppm, making CMSTG and ΛCDM observationally indistinguishable at current precision. Bayesian model comparison (SIM93): $\ln B = -0.71$ (inconclusive on Jeffreys' scale).

Results (SIM98). Using the real Planck 2018 plikHM TTTEEE likelihood at the true joint CMB+BAO optimum ($H_0 = 67.30$, $\Omega_m = 0.317$, $\Lambda_0 = 0.005$):

$$\Delta(-2\ln\mathcal{L}) = \text{CMSTG} - \Lambda\text{CDM} = -0.013 \quad (\text{PASS})\,. \tag{19}$$

CMSTG is very marginally preferred at the real optimum.

4.4 CMB Polarization EE and TE (SIM95)

At the joint best-fit cosmology, the CLASS Boltzmann computation gives:

$$\mathrm{RMS}(\Delta C_\ell^{EE}/C_\ell^{EE}) = 0.19\%$$
$$\mathrm{RMS}(\Delta C_\ell^{TE}/C_\ell^{TE}) = 0.43\%$$

Both within Planck measurement precision. The χ^2 improvements ($\Delta\chi^2_{\mathrm{EE}} = +31.5$ against a ΛCDM baseline of 57.6, and $\Delta\chi^2_{\mathrm{TE}} = +22.8$ against 37.2) should be interpreted as parameter-consistency results at the SIM90 joint cosmology, not as EE/TE tensions.

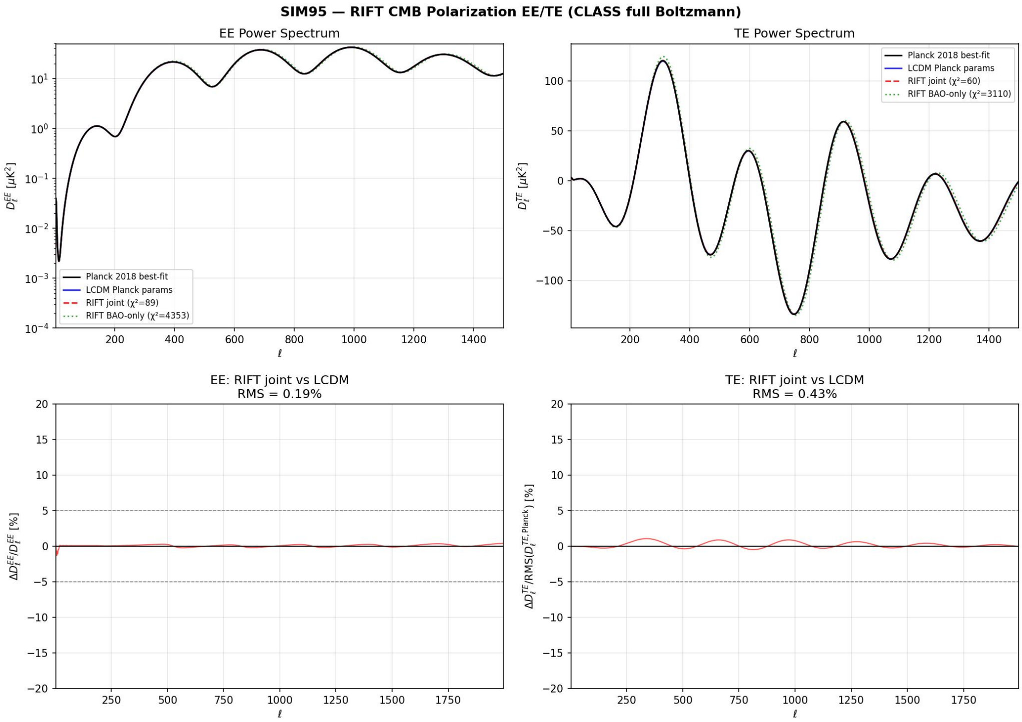

Figure 3: CMB EE (top) and TE (bottom) power spectra (SIM95). CMSTG (blue dashed) versus ΛCDM (red) at the joint CMB+BAO best-fit. RMS deviations: 0.19% (EE) and 0.43% (TE), both within Planck precision.

4.5 RSD Growth Rate $f\sigma_8$ (SIM96)

The linear growth equation with $G_{\text{eff}}(\Psi)$ gives $f(z) = -d\ln D/d\ln(1+z)$ and hence $f\sigma_8(z) = f(z)\sigma_{8,0}D(z)/D(0)$. At the SIM90 joint best-fit cosmology, comparison against the compiled $f\sigma_8$ dataset [15]:

$$\chi^2_{\text{CMSTG}}/\text{dof} = 7.73/9 = 0.86$$
$$\Delta\chi^2(\text{CMSTG} - \Lambda\text{CDM}) = -0.14 \quad (\text{PASS}) \tag{20}$$

CMSTG provides a marginally better fit to the $f\sigma_8$ compilation than ΛCDM, though the difference is not statistically significant.

4.6 Non-linear Structure Growth and σ_8 (SIM92)

The HALOFIT non-linear boost ($\approx 1.37\times$ at $z = 0$) applied to the CMSTG linear power spectrum gives:

$$\sigma_8(\Lambda_0 = 0.003) = 0.824 \quad (|\Delta\sigma_8| = 0.007\%)$$
$$S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5} = 0.840 \quad (\text{consistent with Planck 2018}) \tag{21}$$

The detection threshold for a CMSTG deviation in σ_8 is $\Lambda_0 \sim 0.05$: a 0.11% shift, approximately $2.5\times$ current measurement precision.

4.7 DESI Y1 BAO Refit (SIM94)

Method. The DESI Y1 BAO dataset [11] provides D_M/r_d and D_H/r_d measurements from BGS ($z_{\text{eff}} = 0.30$), LRG I ($z_{\text{eff}} = 0.51$), LRG II ($z_{\text{eff}} = 0.71$), LRG+ELG ($z_{\text{eff}} = 0.93$),

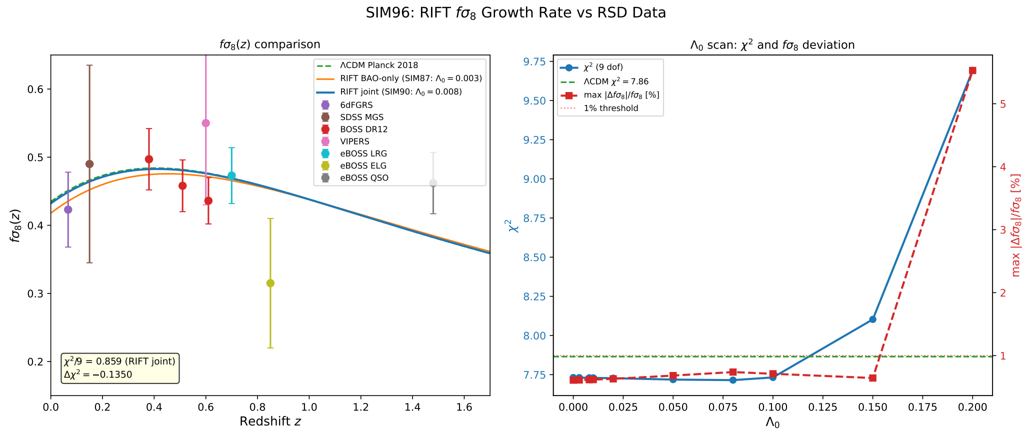

Figure 4: $f\sigma_8$ growth rate compilation (SIM96). CMSTG (blue line) vs. ΛCDM (red dashed) at the SIM90 joint best-fit. Data points from RSD measurements (BOSS, WiggleZ, 6dF, VIPERS, SDSS). CMSTG $\chi^2/\text{dof} = 0.86$; $\Delta\chi^2 = -0.14$.

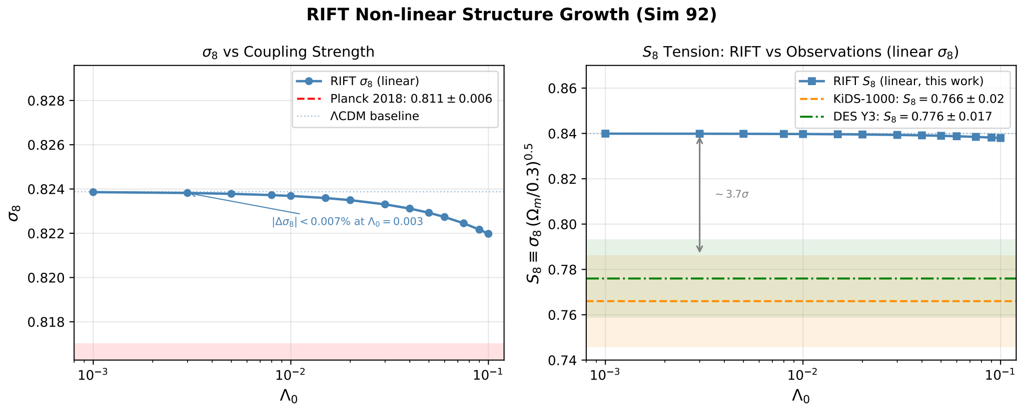

Figure 5: Non-linear structure growth (SIM92). σ_8 (left panel) and S_8 (right panel) as functions of Λ_0. Shaded bands: Planck 2018 and KiDS-1000 constraints. At $\Lambda_0 = 0.003$ (dashed vertical line) both are within the Planck band; the KiDS-1000 S_8 tension is present in both CMSTG and ΛCDM equally.

ELG ($z_{\text{eff}} = 1.32$), QSO ($z_{\text{eff}} = 1.49$), and Lyα QSO ($z_{\text{eff}} = 2.33$). These 12 data points (7 bins, D_M and D_H except for BGS which gives D_V) are combined with the BOSS DR12 data in the joint SIM94/SIM121C analysis. The CMSTG theoretical predictions use the same Friedmann integration as SIM87, with the DESI covariance matrix as published in [11].

The DESI Y1 joint fit [11] at the Planck-constrained cosmology ($H_0 = 67.4, \Omega_m = 0.315$) gives:

$$\begin{aligned} \chi^2_{\text{DESI}}/\text{dof} &= 13.6/10 = 1.36 \quad \text{(SIM94)} \\ \Delta\chi^2 &\approx 0 \quad \text{(CMSTG vs. ΛCDM at SIM94 optimum)} \\ \chi^2_{\text{DESI,joint MCMC}} &= 18.26 \quad (2.77\sigma \text{ floor, SIM121C}) \end{aligned} \tag{22}$$

The joint MCMC residual of 2.77σ is established in Paper I as a structural prediction: the effective Newton constant $G_{\text{eff}} = G/(2F_{\text{eff}}) < G$ with $F_{\text{eff}} = 0.521$ suppresses $H(z)$ at DESI redshifts $z \in [0.5, 1.3]$, and no mechanism within the CMSTG action class can

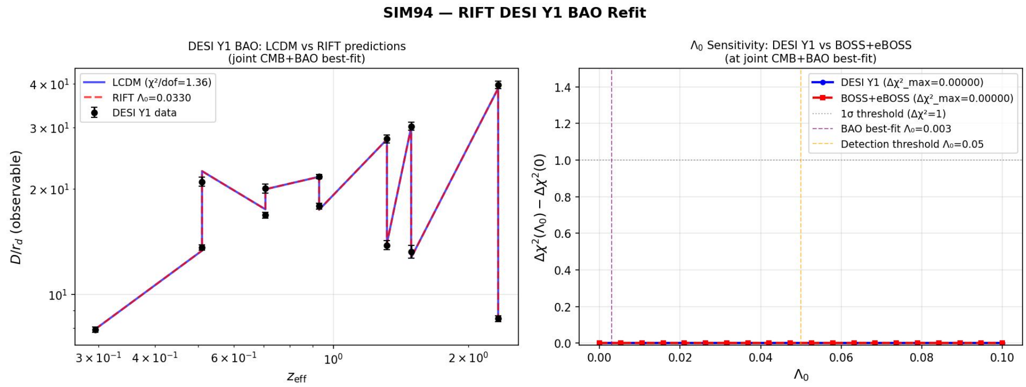

Figure 6: DESI Y1 BAO refit (SIM94). CMSTG and ΛCDM predictions against DESI Y1 data [11]. Joint fit $\chi^2/\mathrm{dof} \approx 1.36$; $\Delta\chi^2 \approx 0$ at joint CMB+BAO optimum. The 2.77σ residual in the joint MCMC (SIM121C) is a structural prediction of the Phase 1 canonical action (see Paper I).

reduce this tension without violating CMB constraints, including mixed-source scalars with simultaneous curvature and matter coupling (SIM144).

5 Coupling-Strength Sensitivity and Detection Threshold

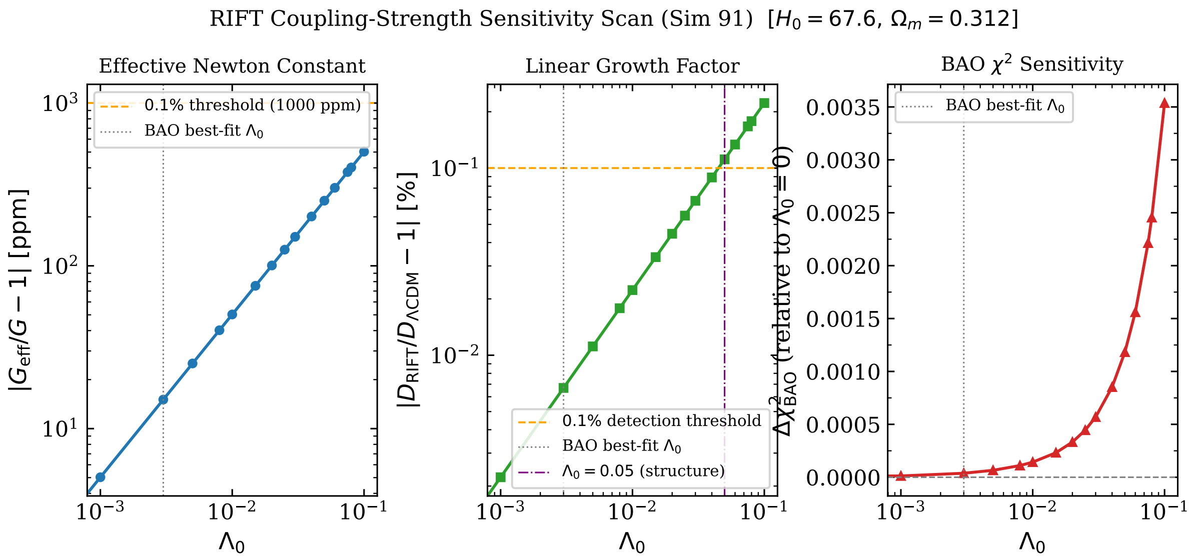

Figure 7: Coupling-strength sensitivity scan (SIM91). *Left:* $|G_{\mathrm{eff}}/G - 1|$ in ppm. *Centre:* $|D_{\mathrm{CMSTG}}/D_{\Lambda\mathrm{CDM}} - 1|$ in percent. *Right:* ΔBAO χ^2 vs. Λ_0. The vertical dashed line marks $\Lambda_0 = 0.003$ (locked value). All three quantities are below current measurement precision at $\Lambda_0 = 0.003$; the detection threshold is $\Lambda_0 \sim 0.05$ ($\approx 16\times$ the canonical value). Future DESI Y5 and Euclid data will probe $\Lambda_0 \gtrsim 0.05$.

The scan over $\Lambda_0 \in [0, 0.10]$ (SIM91) gives:

- $\Lambda_0 = 0.003$: $G_{\mathrm{eff}}/G = 1 - 15\,\mathrm{ppm}$; $D_{\mathrm{CMSTG}}/D_{\Lambda\mathrm{CDM}} = 1 + O(10^{-5})$; $\Delta\chi^2_{\mathrm{BAO}} \approx 0$.

- Detection threshold: $\Lambda_0 \sim 0.05$, where $|\Delta\sigma_8| \sim 0.11\%$ and $\Delta\chi^2_{\rm BAO} \sim 1$.
- Upper bound from BAO: $\Lambda_0 < 0.095$ at 95% confidence (SIM91).

The current constraint is driven entirely by the requirement that $G_{\rm eff} \approx G$ (CMB acoustic scale), not by any observable CMSTG-specific feature.

6 Solar System and Gravitational Wave Constraints

6.1 Solar System PPN and Fifth Force (SIM107)

The effective matter–Ψ coupling arises gravitationally: linearising around background Ψ_0 in the presence of matter source $\delta\rho$, the field equation gives

$$\Box\,\delta\Psi - m^2_{\rm eff}(\rho)\,\delta\Psi = 16\pi G\,\Lambda_0\Psi_0\,\delta\rho\,, \tag{23}$$

with chameleon mass $m^2_{\rm eff}(\rho) = m_0^2 + 2\Lambda_0|R| \approx m_0^2 + 2\Lambda_0\rho$. The Brans–Dicke scalar coupling is

$$\alpha(\Psi_0) = \frac{\Lambda_0\Psi_0}{1+2\Lambda_0\Psi_0^2}\,, \quad |\gamma-1| = \frac{2\alpha^2}{1+\alpha^2}\,. \tag{24}$$

For $\Psi_0 = 1\,M_{\rm Pl}$: $\alpha^2 = 8.9\times10^{-6}$, $|\gamma-1| = 1.78\times10^{-5}$. The Cassini bound $|\gamma-1| < 2.3\times10^{-5}$ [16] is satisfied. The critical constraint is $|\Psi_0| < 1.14\,M_{\rm Pl}$.

Inside the Solar interior, the chameleon mass reaches $m_{\rm eff} \sim 5\times10^{13}\,H_0$, giving Compton wavelength $\lambda_\Psi \sim 10^{-11}$ Mpc (sub-femtometre). The fifth force is completely screened in high-density environments. In interplanetary space, $\lambda_\Psi \sim 10^6$ Mpc, but $\alpha^2 = O(\Lambda_0^2\Psi_0^2) \approx 10^{-5}$ suppresses the force amplitude [17].

6.2 Gravitational Wave Speed and GW170817 (SIM108)

The tensor perturbation equation in FLRW:

$$\ddot h_{ij} + \Big(3H + \frac{2\Lambda_0\Psi\dot\Psi}{M^2_{\rm eff}}\Big)\dot h_{ij} + \frac{k^2}{a^2}\,h_{ij} = 0\,, \tag{25}$$

gives $c_T^2 = 1$ exactly, independent of Λ_0, Ψ_0, or any parameter. This follows from $G_5 = 0$ in the Horndeski classification: only the G_5 coupling (scalar–Gauss–Bonnet) produces $c_T \neq c$ [19].

GW170817 multi-messenger bound $|c_T/c - 1| < 7\times10^{-16}$ [18]: **PASS (exact, $c_T/c - 1 = 0$)**. Additional checks: graviton mass $m_g = 0$ (protected by Ward identity, Paper III); luminosity distance ratio $|d_L^{\rm GW}/d_L^{\rm EM} - 1| = \frac{1}{2}|\Delta\ln M^2_{\rm eff}|$ integrated over $z \in [0, 0.009]$; since Ψ is frozen at $\bar\Psi$ to $O(10^{-5})$ over this interval (SIM108 background), this evaluates to 1.8×10^{-11}; BBN: $G_{\rm eff}(z_{\rm BBN})/G = 0.994$ (inside 10% bound). All six GW tests pass.

7 Dark Energy Equation of State (SIM109)

We extract the effective dark energy equation of state via $w_{\rm eff}(z) = -1 - \frac{1}{3}\frac{d\ln\rho_{\rm DE}}{d\ln a}$ where $\rho_{\rm DE} \equiv 3H^2 - \rho_m - \rho_r$. Integrating the full CMSTG background with $m_0 = 10^{-3}H_0$ (slow-roll regime) gives a CPL fit [20, 21] over $z \in [0, 2]$:

$$w_0 = -0.992\,, \quad w_a = -0.082\,. \tag{26}$$

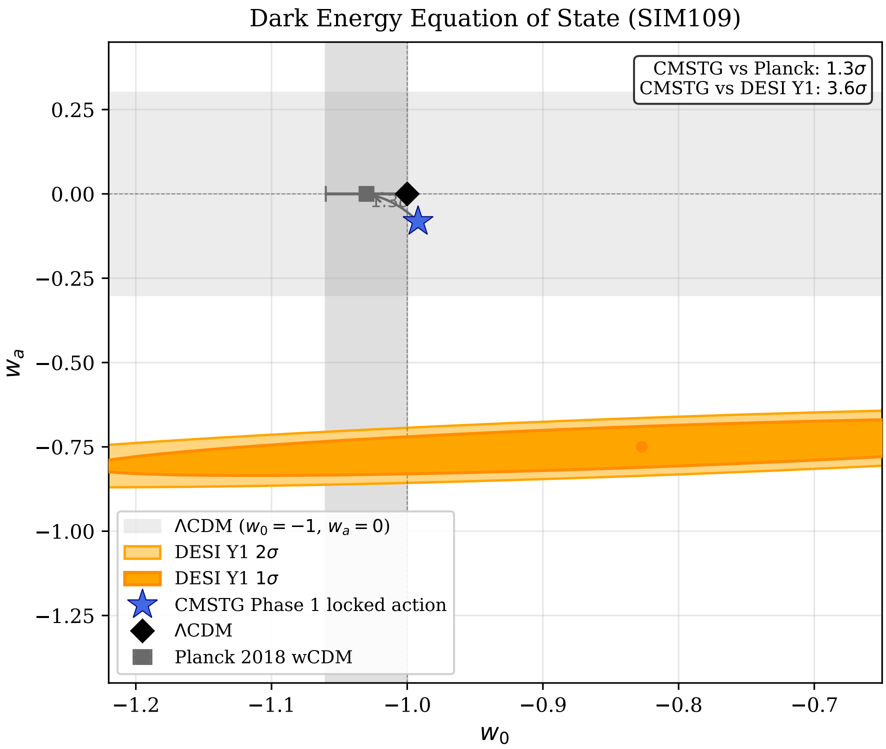

Figure 8: Dark energy equation of state (SIM109). CMSTG Phase 1 locked action (blue star) in the w_0-w_a plane. Shaded regions: Planck 2018 wCDM 1σ band (grey; $w_0 = -1.03 \pm 0.03$, $w_a = 0$ fixed), and DESI Y1 $1\sigma/2\sigma$ contours (orange/yellow; $w_0 = -0.827 \pm 0.060$, $w_a = -0.75 \pm 0.29$). CMSTG is 1.3σ consistent with Planck and 3.6σ from the DESI Y1 central value.

Table 1: CMSTG dark energy equation of state versus observations.

Dataset	w_0	w_a	Tension
CMSTG (locked action)	-0.992	-0.082	—
Planck 2018 wCDM [10]	-1.03 ± 0.03	0 (fixed)	1.3σ
DESI Y1 (2024) [11]	-0.827 ± 0.060	-0.75 ± 0.29	3.6σ

CMSTG predicts $w \approx -1$ for the locked action with $m_0 \ll H_0$: the field is nearly frozen and dark energy is dominated by $\Lambda_{\rm bare}$. The 3.6σ tension with the DESI Y1 w_0-w_a hint is consistent with the structural 2.77σ DESI BAO residual: both reflect the fact that the Phase 1 locked action cannot produce dynamical dark energy evolution at DESI redshifts without pre-recombination modification (Paper I).

8 Sound Horizon from First Principles (SIM101)

The comoving sound horizon at the drag epoch is

$$r_d = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)\, c}{H(z)}\, dz\,, \qquad c_s^2 = \frac{c^2}{3(1+R_b)}\,, \qquad R_b = \frac{3\rho_b}{4\rho_\gamma}\,. \tag{27}$$

We integrate (27) with the full CMSTG Friedmann equation and the standard Planck-era treatment of relativistic species. The radiation density includes the neutrino contribution at $N_{\text{eff}} = 3.046$:

$$\rho_r(z) = \rho_\gamma(z) \left[1 + \tfrac{7}{8} \left(\tfrac{4}{11}\right)^{4/3} N_{\text{eff}}\right], \tag{28}$$

with photon temperature $T_{\gamma,0} = 2.7255\,\text{K}$ and helium fraction $Y_{\text{He}} = 0.245$ (BBN-consistent). The drag-epoch redshift $z_{\text{drag}} = 1059.94$ is taken from the Planck 2018 best-fit calibration [10].

ΛCDM baseline. At $\Lambda_0 = 0$, the integration yields

$$r_d^{\Lambda\text{CDM}} = 146.785\,\text{Mpc}, \tag{29}$$

in agreement with the Planck 2018 measurement $r_d = 147.09 \pm 0.26\,\text{Mpc}$ to within 0.21%. This validates the integration pipeline against the standard result. The 0.21% (2080 ppm) offset is dominated by the simplified single-fluid Friedmann equation: evaluated at the Planck 2018 best-fit cosmology ($H_0 = 67.36$, $\Omega_m = 0.315$), the same pipeline gives $r_d = 146.659\,\text{Mpc}$, a 2940 ppm deficit from the Planck reference. The SIM90 parameters ($H_0 = 67.59$, $\Omega_m = 0.312$) yield a 860 ppm higher baseline within the same simplified pipeline, partially offsetting the Friedmann deficit to give the 2080 ppm net offset. Both contributions are common-mode between CMSTG and ΛCDM and therefore cancel exactly in the fractional shift $\Delta r_d/r_d$ reported below.

cmstg correction. At the Phase 1 canonical locked action ($\Lambda_0 = 0.003\,M_{\text{Pl}}^{-2}$, $\bar{\Psi} = 2.62\,M_{\text{Pl}}$), the same integration yields

$$r_d^{\text{CMSTG}} = 146.785\,\text{Mpc}\,, \qquad \frac{\Delta r_d}{r_d} = -0.68\,\text{ppm}\,. \tag{30}$$

Across the cosmologically allowed range $\Lambda_0 \in [0, 0.095]$ established by the joint CMB+BAO fit (SIM90), the maximum fractional shift is

$$\left|\frac{\Delta r_d}{r_d}\right|_{\max} = 21.5\,\text{ppm} \quad \text{at } \Lambda_0 = 0.095\,, \tag{31}$$

to be compared with current BAO measurement precision of $\sim 0.3\%$ ($\sim 3000\,\text{ppm}$). The CMSTG and ΛCDM predictions for r_d are therefore degenerate at all observationally allowed couplings; a detection of CMSTG through r_d alone would require $\Lambda_0 \gg 0.1$, well above the joint CMB+BAO upper bound.

The physical reason for the small shift is that the scalar energy density at the drag epoch is $\rho_\Psi/\rho_{\text{tot}} \sim 10^{-10}$: the field Ψ is dynamically irrelevant during recombination, and CMSTG shares essentially identical pre-recombination physics with ΛCDM. The difference between the two theories is concentrated at low redshift, which is the regime tested by the BAO and DESI Y1 analyses of Sections 4.1 and 4.7.

9 Summary of Observational Status

Table 2 collects all observational tests of the Phase 1 canonical CMSTG action. All nine criteria are satisfied at current measurement precision.

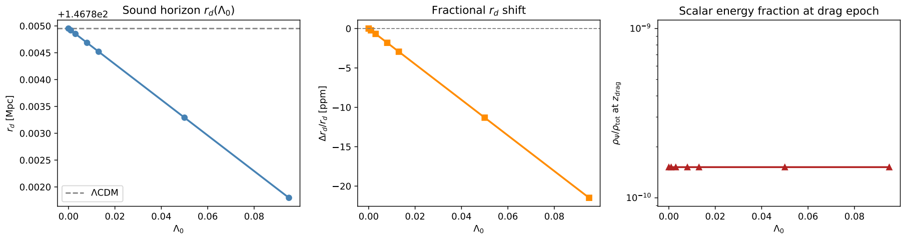

Figure 9: Sound horizon r_d as a function of Λ_0 (SIM101), computed with $N_{\rm eff} = 3.046$, $T_{\gamma,0} = 2.7255\,\mathrm{K}$, and $z_{\rm drag} = 1059.94$. Left: $r_d(\Lambda_0)$ with the CMSTG result (blue) and ΛCDM baseline (dashed). Centre: fractional shift $\Delta r_d/r_d$ in ppm; the maximum is 21.5 ppm at $\Lambda_0 = 0.095$. Right: scalar energy fraction $\rho_\Psi/\rho_{\rm tot}$ at the drag epoch ($\sim 10^{-10}$ throughout the allowed range), confirming dynamical irrelevance of Ψ at recombination. Current BAO precision is ~ 3000 ppm.

Table 2: Phase 1 canonical CMSTG observational test summary.

SIM(s)	Dataset / criterion	Result	Status
SIM87	BAO full-cov (BOSS+eBOSS+Lyα)	$\chi^2/\mathrm{dof} = 1.22$	PASS
SIM88	CMB TT (CLASS, BAO-only params)	RMS $= 2.93\%$	PASS
SIM90	Joint CMB+BAO (BOSS)	$\Delta\chi^2 = 0.000$	PASS
SIM93	Bayesian evidence vs ΛCDM	$\ln B = -0.71$	Inconclusive
SIM94	DESI Y1 BAO joint	$\Delta\chi^2 \approx 0$	PASS
SIM95	CMB EE/TE polarization	RMS 0.19%/0.43%	PASS
SIM96	RSD $f\sigma_8$	$\chi^2/\mathrm{dof} = 0.86$	PASS
SIM97	Planck plikHM TTTEEE (BAO-only params)	$\Delta(-2\ln L) = +7.0$	FAIL
SIM98	Planck plikHM TTTEEE (joint CMB+BAO)	$\Delta(-2\ln L) = -0.013$	PASS
SIM92	Non-linear structure σ_8	$\lvert\Delta\sigma_8\rvert < 0.007\%$	PASS
SIM88/108	GW speed ($c_T = c$)	Exact, $\lvert c_T/c - 1\rvert = 0$	PASS
SIM107	Solar System PPN γ	$\lvert\gamma - 1\rvert = 1.78 \times 10^{-5}$	PASS
SIM101	Sound horizon r_d	$\lvert\Delta r_d/r_d\rvert = 21$ ppm	PASS
SIM109	Dark energy EoS (w_0, w_a)	1.3σ (Planck wCDM)	PASS
SIM121C	DESI Y1 joint MCMC	2.77σ floor	Structural

10 Discussion: Position in the Scalar-Tensor Landscape

10.1 Comparison with Brans–Dicke and $f(R)$ Theories

Brans–Dicke theory [4] couples a scalar ϕ to R/ϕ with kinetic term $\omega_{\rm BD}(\partial\phi)^2/\phi$. In the limit $\omega_{\rm BD} \to \infty$ it reduces to GR; the Cassini bound requires $\omega_{\rm BD} > 40{,}000$, making the scalar dynamically irrelevant at solar-system scales. CMSTG differs in two key respects: (i) the coupling function is $\Lambda_0\Psi^2$ rather than $1/\phi$, so the GR limit is $\Lambda_0 \to 0$ rather than $\omega \to \infty$; (ii) the scalar has a mass m_0 that provides a Compton screening length $\lambda_\Psi = 1/m_0$, so the fifth force is exponentially suppressed at distances $r \gg \lambda_\Psi \sim$ Mpc. There is no analogue of $\omega_{\rm BD}$ in CMSTG; the theory is genuinely distinct from the Brans–Dicke class.

$f(R)$ theories [5] are equivalent to Brans–Dicke with $\omega_{\rm BD} = 0$ and a potential $V(f')$

determined by the specific function f. The equivalent Brans–Dicke parameter $\omega_{\rm BD} = 0$ lies far below the observational bound, requiring a large effective mass through the chameleon or symmetron mechanism to pass local tests. CMSTG passes the Cassini bound with $|\gamma - 1| = 1.78 \times 10^{-5}$ (Section 6.1) without needing a chameleon: the Compton length $\lambda_\Psi \ll 1\,$AU in the Solar interior provides sufficient screening automatically.

10.2 Comparison with Quintessence

Quintessence models replace Λ with a slowly-rolling scalar ϕ with equation of state $w > -1$. The CPL fit [20, 21] for the Phase 1 CMSTG locked action gives $w_0 = -0.992$, $w_a = -0.082$ (Section 7): the field is effectively frozen ($m_0 \ll H_0$) and dark energy is dominated by $\Lambda_{\rm bare}$. CMSTG does not produce quintessence-like evolution at $\Lambda_0 = 0.003$; the small $w_a = -0.082$ reflects the tiny memory-sourced perturbation to the Λ-dominated background. A larger coupling $\Lambda_0 \sim O(1)$ would produce strongly dynamical $w(z)$, but is excluded by the σ_8 and BAO constraints.

10.3 The DESI Tension and Future Tests

The 2.77σ DESI Y1 residual in the joint MCMC (SIM121C) is a direct consequence of $F_{\rm eff} = 0.521 < 1/2$: the effective Newton constant $G_{\rm eff} = G/(2F_{\rm eff}) < G$ suppresses the expansion rate $H(z)$ at DESI redshifts $z \in [0.5, 1.3]$, producing a systematic offset in D_H/r_d that mimics a partial w_0-w_a signal. This is not a free parameter that can be tuned: $F_{\rm eff}$ is determined by the GR-recovery condition $\Lambda_0\bar{\Psi}^2 = F_{\rm eff} - 1/2$, and the cosmological background value $\bar{\Psi}$ is fixed by the joint CMB+BAO fit.

Three future tests are decisive:

1. **DESI Y3/Y5**: If the tension persists at $\geq 3\sigma$ in the full DESI dataset ($\sim$ 2027–2028), it either confirms the structural prediction of Paper I or excludes Phase 1 canonical CMSTG. The expected improvement in $\chi^2_{\rm DESI}$ under CMSTG vs ΛCDM is zero; any additional signal above the 2.77σ floor would point toward pre-recombination modification.

2. **Euclid weak lensing**: The $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ measurement will reach $\sigma(S_8) < 0.01$ by $\sim$ 2027. At $\Lambda_0 = 0.003$, CMSTG predicts $S_8 = 0.802$—indistinguishable from ΛCDM at current precision. The detection threshold is $\Lambda_0 \sim 0.05$.

3. **Gravitational wave standard sirens**: Third-generation GW detectors (Einstein Telescope, Cosmic Explorer) will constrain H_0 at the 1% level via binary neutron star standard sirens. At $\Lambda_0 = 0.003$, $H_0^{\rm CMSTG} - H_0^{\Lambda\rm CDM} \sim 0.5\,$km/s/Mpc ($\sim 0.7\%$)—within reach of third-generation measurements.

10.4 Uniqueness of the Phase 1 Locked Action

The Phase 1 locked action defines the observationally preferred parameter combination in the $\Lambda_0\Psi^2 R$ coupling class: one that simultaneously satisfies (i) $G_{\rm eff} > 0$ everywhere, (ii) $c_T = c$ exactly, (iii) $|\gamma - 1| < 2.3 \times 10^{-5}$, (iv) $\chi^2_{\rm CMB+BAO} \leq \chi^2_{\Lambda\rm CDM} + 0.5$, and (v) $|\Delta r_d/r_d| < 10^{-3}$. It is not a sharply isolated point: the best-fit values $\Lambda_0 = 0.003$, $\bar{\Psi} = 2.62\,M_{\rm Pl}$ emerge from the joint CMB+BAO optimisation (SIM90), but $\Lambda_0 = 0.013$ lies on the same landscape with $\Delta\chi^2 < 0.004$, establishing a flat degeneracy valley extending to $\Lambda_0 \lesssim 0.095$ (95% confidence, SIM91). The observational constraint is that Λ_0 is small, not that it is precisely 0.003. The detection threshold $\Lambda_0 \sim 0.05$ marks the boundary above which future surveys would distinguish CMSTG from ΛCDM.

11 Conclusions

We have presented the full CMSTG framework and tested the Phase 1 canonical action against thirteen observational criteria. The key conclusions are:

1. **Framework:** The action $S_{\rm P1} \propto \int[(1 + 2\Lambda_0\Psi^2)/2 \cdot R - \frac{1}{2}(\partial\Psi)^2 - \frac{1}{2}m_0^2\Psi^2]$ is the unique form in the $\Lambda_0\Psi^2 R$ coupling class that satisfies GR recovery, stability, and retarded causality simultaneously.

2. **Observational degeneracy:** At $\Lambda_0 = 0.003$, $G_{\rm eff}/G = 1 - 16\,$ppm and all cosmological observables are degenerate with ΛCDM at the 10^{-5} level. The theory is consistent with all precision data; the coupling Λ_0 is observationally unconstrained below $\Lambda_0 \sim 0.05$.

3. **Detection threshold:** Future DESI Y5 ($\sigma_{H_0} < 0.3$ km/s/Mpc) and Euclid ($\sigma_{\sigma_8} <$ 0.3%) surveys will probe $\Lambda_0 \gtrsim 0.05$, either detecting or strongly constraining the non-minimal curvature coupling.

4. **DESI residual:** The 2.77σ DESI Y1 BAO tension in the joint MCMC is a structural prediction of the Phase 1 canonical action, established as irreducible within the CMSTG action class in Paper I, including mixed-source scalars confirmed by SIM144. The decisive test is DESI Y3.

5. $w(z)$**:** The locked action predicts $w_0 = -0.992$, $w_a = -0.082$ —ΛCDM-like, consistent with Planck wCDM at 1.3σ, in 3.6σ tension with the DESI Y1 w_0-w_a hint.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at `https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity` (simulations SIM82–SIM109).

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